# SumOfSquares: A Julia package for Polynomial Optimization

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# History

- Started in August 2016: SOSTOOLS demos.
- October 2016: Amelia Perry, Joey Huchette, Robin Deits and Chris Coey join, package split into `MultivariatePolynomials`, `PolyJuMP` and `SumOfSquares`.
- April 2017, `MultivariatePolynomials` split into `MultivariatePolynomials`, `DynamicPolynomials`, `TypedPolynomials`,
- August 2017 `MultivariateMoments` and `SemialgebraicSets` moved out of `SumOfSquares`.
- June 2017–March 2019 Development of `MathOptInterface` (MOI) → rewrite of JuMP, `PolyJuMP` and `SumOfSquares`.
- July 2019–December 2020 Development of `MutableArithmetics` for performance.
- 2019–2020 `MomentOpt` and Tillmann Tweisser push for more features.

# MutableArithmetics

What is mutable in (JuMP/MOI var: $\alpha$, Polynomial var: $x$):

$$A = \begin{bmatrix} \alpha x + 1 & x - \alpha \\ x - \alpha & x + 2 \end{bmatrix}$$

# MutableArithmetics

What is mutable in (JuMP/MOI var: $\alpha$, Polynomial var: $x$):

$$A = \begin{bmatrix} \alpha x + 1 & x - \alpha \\ x - \alpha & x + 2 \end{bmatrix}$$

3 nested levels of mutability (4 with e.g. `BigInt`/`BigFloat`) $\rightarrow$ huge toll on performance.

MutableArithmetics API:

- `MA.operate!` / `MA.operate_to!`: May mutate first argument.
- `MA.mutable_operate!` / `MA.mutable_operate_to!`: Must mutate first argument.

# Polynomial basis

$$p(x) = b(x)^\top Q b(x)$$

Basis $b$ from `MultivariateBases` package:

- `MonomialBasis`: $x^2, xy, y^2$
- `ScaledMonomialBasis`: $x^2, \sqrt{2}xy, y^2$
- Orthogonal basis: Hermite, Laguerre, Legendre and Chebyshev bases.
- More to come... (e.g. Lagrange interpolation basis)

# Certificates

*Ideal* certificates (SOS on algebraic variety):

- Remainder: $p(x)$ SOS when $x \in S \rightarrow$ remainder modulo $S$.
- Newton polytope.
- Maxdegree (all monomials up to degree $d$).

*Preorder* certificates (SOS on basic semialgebraic set):

- Putinar certificate:

$$g_i(x) \geq 0 \Rightarrow p(x) \geq 0 \quad \leftrightarrow \quad p(x) = s_0(x) + \sum_i s_i g_i.$$

Flexibility by combining existing or creating new certificates.

# Sparsity reduction

$$p(x) = b(x)^\top Q b(x) \to p(x) = b_1(x)^\top Q_1 b_1(x) + \cdots + b_k(x)^\top Q_k b_k(x)$$

- `SignSymmetry` certificate.
- Chordal sparsity certificates:
  - `VariableSparsity`: Variable/correlative sparsity.
  - `MonomialSparsity`: Monomial/term sparsity.

  Chordal *extension* implementation with *greedy* fill-in heuristics. Should move to separate package (with `COSMO` ?).

# Sums of Hermitian squares / non-commutative

$$p(x) = b(x)^\top Q b(x)$$

with $x_1 x_2 \neq x_2 x_1$, $Q \in \mathbb{C}^{n \times n}$ hermitian PSD:

- Initial work on extending MOI to *complex* numbers in `ComplexOptInterface`, e.g.

  complex hermitian PSD $\rightarrow$ real symmetric PSD.

- *Non-commutative* algebra implemented in `DynamicPolynomials`, i.e. created with `@ncpolyvar`.
- Implementation of Newton polytope for non-commutative variables.
- Still features missing compared to `NCSOStools`, e.g. atom extraction with `MultivariateMoments`.

# Future work

- Polynomial basis for equality (Pull Requests opened by Tillmann Weisser).
- Symmetry reduction (Good progress made by Tillmann Weisser and Marek Kaluba).
- Certificates for `PolyJuMP`: equality in ideal without Groebner basis.
- Better integration with `Hypatia` (just released!)
- More certificates: Schmüdgen, ...
- Rework examples with Literate to include in documentation.